# A Lightweight Hybrid Retrieval-Augmented Generation System for Chinese Course Materials

**Track:** B — Demonstration
**Submission Date:** October 12, 2026

## 1. Team Members and Responsibilities

| Member | Individual Technical Responsibility |
|---|---|
| Li Ranjing | Document extraction, chunking algorithms, and chunk-size ablation experiments. |
| Ma Haidong | BM25 sparse retrieval, indexing, and lexical baseline experiments. |
| Wang Xinyi | BGE dense retrieval, vector indexing, and semantic retrieval experiments. |
| Ma Siyuan | Implementation of reciprocal rank fusion (RRF), reranking, and retrieval ablation experiments. |
| Li Haotian | Evidence-grounded answer generation, citation verification, user interface development, and end-to-end testing |

## 2. Background and Motivation

University students frequently search through lecture slides, course notes, and PDF readings to locate precise explanations. Traditional keyword-based retrieval methods may fail to recognize semantically relevant information expressed using different terminology. Meanwhile, standalone large language models (LLMs) may generate inaccurate answers without reliable supporting evidence.

Retrieval-Augmented Generation (RAG) combines external document retrieval with language generation to improve the accessibility and reliability of knowledge-intensive question answering [1].

This project aims to develop a lightweight RAG-based question-answering system for Chinese course materials. Rather than training a large language model from scratch, we will focus on hybrid retrieval, ranking optimization, and evidence-grounded answer generation, making the system practical under limited computational resources.

## 3. Problem Definition and Research Questions

The objective is to design and implement a system that answers natural-language questions based on a collection of course documents.

Given a user query, the system should retrieve relevant document passages, generate an answer supported by the retrieved evidence, and provide traceable source citations. When sufficient evidence cannot be found, the system should indicate uncertainty or abstain from answering.

The project addresses three research questions:

**RQ1:** Does hybrid retrieval combining sparse and dense methods outperform individual retrieval approaches?

**RQ2:** How do document chunking strategies and reranking affect retrieval accuracy and computational efficiency?

**RQ3:** How can retrieved evidence and citation verification improve answer reliability and the handling of unanswerable questions?

# 4. Proposed System and Technical Approach

The proposed system consists of five major components.

**Document Processing:** Uploaded PDF documents and lecture materials will be parsed, cleaned, and divided into smaller text chunks. Different chunking strategies will be investigated to determine their influence on retrieval performance.

**Sparse and Dense Retrieval:** BM25 will be implemented as a lexical retrieval baseline. A lightweight Chinese embedding model, BAAI/bge-small-zh-v1.5 [3], will be used for semantic retrieval, with FAISS supporting vector indexing and similarity search.

**Hybrid Retrieval and Reranking:** We will implement reciprocal rank fusion (RRF) [2] to combine BM25 and dense retrieval results. A lightweight pretrained reranking model, such as BAAI/bge-reranker-base, will be evaluated for refining the ranking of retrieved passages. If computational resources are limited, reranking will be treated as an optional enhancement.

**Evidence-Grounded Answer Generation:** Retrieved passages will be incorporated into structured prompts for answer generation using a fixed pretrained language model. The system will display source document names and relevant passages to support answer traceability. When the retrieved evidence is insufficient, the model will be instructed to abstain from answering. A lightweight local model or an external inference API will be selected based on available computational resources, and the same model will be used across all experimental configurations.

**Interactive Demonstration:** A Streamlit-based web interface will support document uploading, question answering, source inspection, and retrieval-method comparison.

The main independent contributions will be the implementation and evaluation of document processing strategies, retrieval modules, ranking fusion, and evidence verification. Existing pretrained models and open-source libraries will be explicitly acknowledged.

# 5. Experimental Design and Evaluation

We plan to construct an evaluation dataset from authorized Chinese course materials, containing approximately 100–150 manually annotated questions. Each answerable question will be associated with a reference answer and supporting document passages. The dataset will also include unanswerable questions to evaluate the system's ability to recognize insufficient evidence.

Four retrieval configurations will be compared:

| Configuration | Description |
|---|---|
| B1 | BM25 sparse retrieval |
| B2 | BGE-based dense retrieval |

| Configuration | Description |
|---|---|
| H1 | Hybrid retrieval using BM25, BGE, and RRF |
| H2 | Hybrid retrieval using BM25, BGE, RRF, and reranking |

**Retrieval Evaluation:** Recall@5 and MRR@10 will be used to evaluate retrieval effectiveness, while response latency will measure computational efficiency.

**Answer Evaluation:** Answer correctness and citation support will be assessed using a predefined human evaluation rubric. For unanswerable questions, we will measure the proportion of cases in which the system correctly abstains from providing an unsupported answer.

**Ablation Studies:** We will investigate the effects of document chunk size, hybrid fusion, and reranking on retrieval performance.

To ensure fair comparisons, all retrieval configurations will use the same document collection, evaluation questions, generation model, and prompt configuration. Experimental results will be analyzed to identify the advantages and limitations of each approach.

# 6. Expected Outcomes

The expected deliverables include:

1. A functional web-based RAG demonstration supporting document uploading, question answering, and source citations.
2. Independently implemented retrieval and fusion modules.
3. Experimental comparisons of sparse, dense, hybrid, and reranked retrieval.
4. An annotated evaluation dataset and reproducible evaluation scripts.
5. A complete GitHub repository containing source code, environment requirements, documentation, and individual contribution records.
6. A final technical report following the ACL paper format, presentation slides, and a demonstration video.

We expect the project to provide practical insights into the effectiveness and efficiency of hybrid retrieval in educational question-answering applications. Improvements will be determined through experimental evaluation rather than assumed in advance.

# 7. Project Timeline

| Period | Planned Activities |
|---|---|
| October 12, 2026 | Submit the project proposal and finalize individual responsibilities. |
| Weeks 1–2 | Collect documents, define annotation procedures, and implement document processing, BM25, and dense retrieval baselines. |
| Weeks 3–4 | Implement hybrid fusion, reranking, answer generation, and the initial interactive prototype. |

| Period | Planned Activities |
| --- | --- |
| Weeks 5–6 | Conduct comparative and ablation experiments, analyze errors, optimize the demonstration, and prepare final materials. |
| Official deadlines TBD | Submit the midterm and final deliverables according to the course schedule. |

# References


[1] Lewis, P., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. Advances in Neural Information Processing Systems (NeurIPS). https://arxiv.org/abs/2005.11401

[2] Cormack, G. V., Clarke, C. L. A., and Buettcher, S. (2009). *Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods*. Proceedings of SIGIR, 758–759. https://doi.org/10.1145/1571941.1572114

[3] Xiao, S., Liu, Z., Zhang, P., and Muennighoff, N. (2023). *C-Pack: Packaged Resources To Advance General Chinese Embedding*. arXiv:2309.07597. https://arxiv.org/abs/2309.07597